Homework 2: Flow Matching & DDIM Implementation

CMU 10-799: Diffusion & Flow Matching

Spring 2026

Due: Tuesday, February 3, 2026 at 11:59 PM ET **Total Points:** 100

Late Due Date: Thursday, February 5, 2026 at 11:59 PM ET

Submission: Gradescope https://www.gradescope.com/courses/1207241

Starter Code: https://github.com/KellyYutongHe/cmu-10799-diffusion/

Introduction

Welcome to Homework 2! In HW1, you built a DDPM [Ho et al., 2020] model that learns to denoise images step by step. Now you'll **implement Flow Matching** [Lipman et al., 2023], a simpler and often more efficient framework that learns a velocity field to transport noise to data along straight paths.

In addition to flow matching, you will also **implement DDIM** [Song et al., 2020], a deterministic sampler that lets your existing DDPM model generate images in far fewer steps. Explore and compare flow matching and DDIM with DDPM sampling from HW1!

By the end of this homework, you'll have two working generative models (DDPM and Flow Matching) and will **choose your specialization track** for the rest of the course.

This homework is AI-friendly. Same rules as HW1. You may use any AI coding assistants, chatbots, or reference implementations. You may also use any other resources that you can find on the Internet. At the end of the homework, you'll document what resources you used.

Part 0: Setup your codebase (0 point)

You'll continue using the same codebase from HW1. Your DDPM implementation should still work and you'll need it for this homework. **What You'll Implement:**

File	What to implement
`src/methods/flow_matching.py`	Create your own flow matching implementation
`src/methods/ddpm.py`	Add DDIM sampling
`configs/`	Add configs for flow matching & DDIM
`scripts/`	Add scripts for flow matching & DDIM
`train.py`	Add flow matching training options
`sample.py`	Add flow matching & DDIM sampling options

Hint for Flow Matching:

Your `flow_matching.py` should follow the same structure as your `ddpm.py`:

- A `FlowMatching` class that inherits from `BaseMethod`
- `compute_loss(x_0, ...)` — returns training loss given a batch of real images
- `sample(batch_size, image_shape, ...)` — generates new images starting from noise

The core algorithm is different (and simpler!) but the interface should feel familiar.

Compute Budget:

You have **$500 in Modal credits** for the entire course (all 4 homework). Budget wisely!

Part I: Implement Flow Matching (25 points)

Now it's time to build your flow matching model! Take a good look at what you have built so far and start building again! Remember, you are free to add, delete and modify any and all parts of the starter code to fit your preference.

Q1. Building Intuition [optional, 0 pts]

You can still leverage the same strategies from HW1 to develop some intuitions of the algorithm with some toy experiments before diving into the full scale training.

(a) [0 pts] You can still use the Jupyter notebook `01_1d_playground.ipynb` to experiment with Flow Matching on 1D toy data. This notebook contains some options of mixture of Gaussians and their visualizations, you can try out your algorithm first in this playground.

(b) [0 pts] As with HW1, you can use the `--overfit-single-batch` flag to sanity check your implementation before full-scale training.

Q2. Implementation [25 pts]

Now let's implement flow matching! Use the same codebase you have from HW1 as the skeleton, and your job is to fill in the core algorithm. Remember: this is your codebase. Feel free to modify any part of the starter code to fit your preferences. Add helper functions, reorganize files, change the config structure... whatever helps. The only requirement is that your final code runs and produces good results.

(a) [5 pts] Train your flow matching model to convergence. You may use either the provided configs or your own. **You should use roughly the same model architecture, training iterations and batch size as your DDPM model.** Report:

1. Model size
2. Batch size and total training iterations
3. Training loss curve
4. Compute cost (GPU hours)
5. Number of sampling steps

- **Model Size:** The U-Net model has approximately **19.85M** parameters.
- **Training Configuration:** We trained for a total of **120,000** iterations with a batch size of **128**.
- **Compute Cost:** The training took approximately **6** hours on a single NVIDIA L40S GPU.
- **Sampling Steps:** 100

Training Loss Curve: The model converged stably. The training loss curve is shown below:

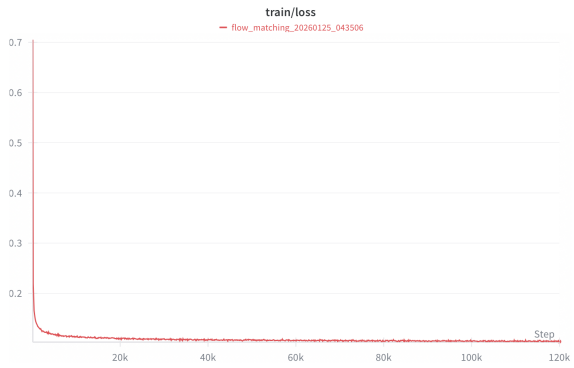
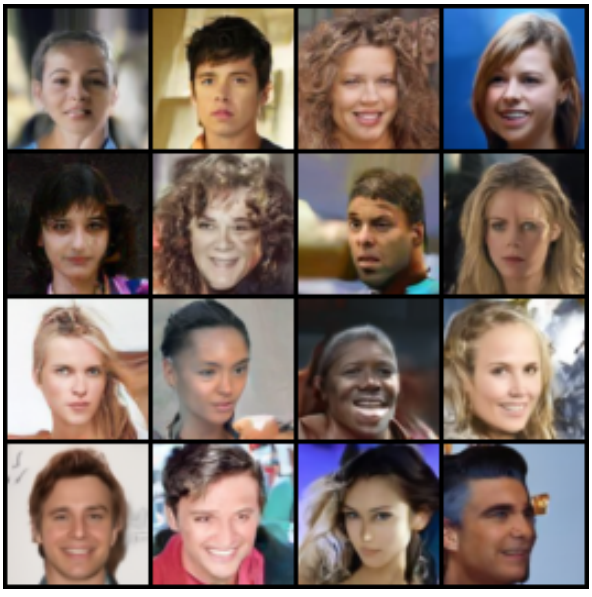


Figure 1: Training loss curve for Flow Matching over 120k iterations.

(b) [10 pts] Generate a grid of 16 samples from your trained model. Include this grid below.



(c) [10 pts] Evaluate your model KID with 1k samples and report your KID score (both mean and

std). You should be able to obtain KID < 0.005 with single L40S GPU training for a few hours.

frechet_inception_distance: 19.83889
kernel_inception_distance_mean: 0.003977416
kernel_inception_distance_std: 0.0002603464

(d) [0 pts] Provide a zip file of your code through Gradescope "Homework 2 Code" assignment. Make sure your code is runnable because we may run it to verify your results, and do not include any large files such as model checkpoints or dataset files. If you do not provide a zip file for your code, you will receive 0 point on Part I, Part III and Part IV of this homework.

Part II: Help Your "Classmate" Debug (15 points)

Kale is back! She's trying Flow Matching now but running into new issues, yet again. In each of the following scenario, you may be provided a loss curve, a grid of samples or a description of the situation. Try your best to help Kale with her implementation!

Q3. Pseudo Debugging Scenarios [15 pts]

(a) [5 pts] Kale trained both DDPM (from HW1) and flow matching on the same dataset with the same model architecture. She notices that her converged Flow Matching loss is significantly higher than her converged DDPM loss. Worried that something is wrong, she asks you: is this expected? Should she be concerned?

Yes, this is expected! Kale should not worry. The reason for the discrepancy is that she is optimizing two models with different objectives and different scales, so the raw losses are not directly comparable. In standard DDPM, the model is trained to predict the added noise, ϵ. Since $\epsilon \sim \mathcal{N}(0, I)$, the target has a variance of roughly 1. In Flow Matching (specifically the Optimal Transport/Straight Path formulation), the model predicts the velocity field v_t, which points from the source (noise) to the target (data). The target is typically $x_{data} - x_{noise}$. Since the noise and data are roughly independent, the variance of this difference is approximately the sum of their variances. Because the variance of the Flow Matching target $(x_{data} - x_{noise})$ is larger than the DDPM target (ϵ), the Mean Squared Error will naturally be higher for Flow Matching, even if both models are learning equally well. Comparing the absolute loss values between two different loss functions is meaningless.

(b) [5 pts] Kale's training loss starts reasonable but occasionally spikes to very large values or NaN, especially early in training. Her DDPM implementation worked fine with the same hyperparameters and learning rate. What might be the cause and how to fix it? Identify the issue and provide two potential strategies to fix the bug.

In DDPM, the model tries to predict ϵ, which is standard normal (mean 0, variance 1). In Flow Matching, the model tries to predict the velocity $v_t = x_1 - x_0$. This target is unscaled. If your data values or noise values are large, the target vector can be huge. Because the target values are larger, the gradients calculated during backpropagation are significantly

larger than in DDPM. Using the same learning rate means you are taking much bigger steps in the weight space. Early in training, when the model is wrong and the loss is highest, a single "bad batch" with large vectors can produce a massive gradient update that pushes the model weights to infinity (NaN).
Methods to fix:

1. Gradient Clipping: This is the most direct fix. By capping the maximum norm of the gradient vector before the optimizer step, you prevent rare exploding batches from destroying the model weights.

2. Learning Rate Warmup: Since the Flow Matching loss landscape is steeper, starting immediately with the high learning rate used for DDPM is risky. Linearly increase the learning rate from 0 to the target value over the first few thousand iterations. This allows the model to settle into a stable region of the loss landscape with small, safe steps before accelerating the training speed.

(c) [5 pts] After the model fully converged, Kale tried to sample from her model and obtained the images below. She's using 200 Euler steps and her training loss looks normal. What are the debugging steps that you would recommend her to take?

Figure 2: The samples that Kale obtained.

The high-frequency noise suggests a mismatch in the sampling logic. Kale should check:

1. **Check Integration Direction:** Ensure the Euler loop integrates *forward* from $t = 0$ (noise) to $t = 1$ (data). If she copied the DDPM logic ($T \to 0$), she is integrating in the wrong direction.

2. **Check Time Input:** Verify the model receives continuous time $t \in [0, 1]$ (e.g., 0.5)

rather than integer indices (e.g., 500). Integer inputs will cause the sinusoidal embeddings to output high-frequency noise.

3. **Enable Eval Mode:** Ensure `model.eval()` is called before sampling to disable stochastic behavior in Dropout/BatchNorm layers which can cause drift in the ODE solve.

4. **Unnormalization:** Confirm the output is unnormalized from $[-1, 1]$ to $[0, 1]$ before saving. Otherwise, negative values are clipped to black, causing contrast artifacts.

Part III: Implement DDIM (20 points)

So far you've trained two generative models: DDPM (HW1) and Flow Matching (this homework). You may have noticed that DDPM requires many more sampling steps (∼1000) compared to Flow Matching (∼100) to produce good results. Can we make DDPM faster without retraining?

DDIM (Denoising Diffusion Implicit Models) [Song et al., 2020] answers this question. The key insight is that DDPM's reverse process can be generalized to a family of non-Markovian processes that share the same training objective. By setting the stochasticity to zero, we get a deterministic ODE that can be solved with far fewer steps — using the exact same trained model.

The DDIM update rule is:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \underbrace{\left(\frac{x_t - \sqrt{1-\bar{\alpha}_t} \cdot \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \right)}_{\hat{x}_0} + \sqrt{1-\bar{\alpha}_{t-1}} \cdot \epsilon_\theta(x_t, t) \tag{1}$$

Or in pseudocode:

Algorithm 1 DDIM Sampling

Require: trained noise predictor ϵ_θ, number of steps S, noise schedules $\bar{\alpha}$

Sample $x_T \sim \mathcal{N}(0, I)$
Create timestep subsequence $[\tau_S, \tau_{S-1}, ..., \tau_1]$ from $[T, ..., 1]$ ▷ e.g., $[1000, 900, 800, ...]$
for $i = S, S-1, ..., 1$ **do**
    $t \leftarrow \tau_i$
    $t_{\text{prev}} \leftarrow \tau_{i-1}$ (or 0 if $i = 1$)
    $\epsilon \leftarrow \epsilon_\theta(x_t, t)$ ▷ Predict noise using your trained DDPM
    $\hat{x}_0 \leftarrow \frac{x_t - \sqrt{1-\bar{\alpha}_t} \cdot \epsilon}{\sqrt{\bar{\alpha}_t}}$ ▷ Predict clean image
    $x_{t_{\text{prev}}} \leftarrow \sqrt{\bar{\alpha}_{t_{\text{prev}}}} \cdot \hat{x}_0 + \sqrt{1-\bar{\alpha}_{t_{\text{prev}}}} \cdot \epsilon$ ▷ DDIM step
end for
return x_0

Q4. DDIM Implementation [20 pts]

Implement DDIM sampling for DDPM model that you trained for HW1. No retraining needed! DDIM works with your existing trained model!

(a) [10 pts] Generate a grid of 16 samples using 100 DDIM steps and include it below.

(b) [10 pts] Evaluate KID with 1k samples using 100 DDIM steps. Report your KID score (mean and std). How does this compare to your original DDPM with 1000 steps and your Flow Matching model?

kernel_inception_distance_mean: 0.006804693
kernel_inception_distance_std: 0.0003628672
The score is not as good as the original DDPM score, but close enough. The main boost was in sampling time. Also, when compared to flow matching, DDIM looses once again. Flow matching provides a much better metric score with a similar sampling time.

Part IV: Ablation Study (25 points)

In this part of the homework, we will explore how the number of sampling steps affects sample quality for both Flow Matching and DDIM.

Q5. Sampling Steps Ablation [25 pts]

You now have three ways to generate images: DDPM (1000 steps from HW1), DDIM (fewer steps, same trained DDPM model), and Flow Matching (Euler integration). Let's see how they compare.

(a) [15 pts] Report KID scores (mean and std) for Flow Matching and DDIM at different step counts. Fill in the table below:

Steps	Flow Matching KID	DDIM KID
1	$\mu = 0.2641, \sigma = 0.0019$	$\mu = 0.6398, \sigma = 0.0020$
5	$\mu = 0.0347, \sigma = 0.0009$	$\mu = 0.0271, \sigma = 0.0008$
10	$\mu = 0.0174, \sigma = 0.0006$	$\mu = 0.0147, \sigma = 0.0005$
50	$\mu = 0.0057, \sigma = 0.0003$	$\mu = 0.0084, \sigma = 0.0004$
100	$\mu = 0.0042, \sigma = 0.0003$	$\mu = 0.0069, \sigma = 0.0004$
200	$\mu = 0.0038, \sigma = 0.0003$	$\mu = 0.0041, \sigma = 0.0003$
1000	$\mu = 0.0027, \sigma = 0.0002$	$\mu = 0.0020, \sigma = 0.0002$

(b) [10 pts] Based on your results, compare flow matching, DDIM, and your original DDPM from HW1:

1. At what step count does each method start producing reasonable samples?
2. For each method, what is the minimum number of steps needed to achieve similar quality to your DDPM with 1000 steps from HW1?
3. How do you compare flow matching and DDIM with DDPM at 100 steps? What about at 1000 steps?
4. If you wanted the best sample quality regardless of speed, which method would you choose? If you wanted fast generation with acceptable quality, which would you choose?

Both Flow Matching and DDIM start producing recognizable, reasonable samples as early as **10 steps**.

- **DDIM** achieves a KID of 0.0147 at 10 steps.
- **Flow Matching** achieves a KID of 0.0174 at 10 steps.

In contrast, standard DDPM fails completely at these low step counts, producing unrecognizable noise even at 100 steps (KID ≈ 0.6994).

For each method, what is the minimum number of steps needed to achieve similar quality to your DDPM with 1000 steps from HW1?
The baseline DDPM at 1000 steps achieved a KID of ≈ 0.0047.

- **Flow Matching:** Matches this quality at **100 steps** (KID = 0.0042). Even at 50 steps, it is very close (0.0057).
- **DDIM:** Matches this quality at **200 steps** (KID = 0.0041). At 100 steps, it is slightly worse (0.0069) but visually comparable.

How do you compare flow matching and DDIM with DDPM at 100 steps? What about at 1000 steps?
At 100 Steps: Standard DDPM collapses completely (KID ≈ 0.70), producing useless noise. Both accelerated methods thrive here: Flow Matching (0.0042) outperforms DDIM (0.0069) slightly, likely because the straight-path ODE is numerically easier to integrate than the diffusion ODE at this resolution.
At 1000 Steps: All methods perform well. DDIM achieves the best score (0.0020), followed closely by Flow Matching (0.0027) and standard DDPM (0.0047). The gap between them is

small, but DDIM and Flow Matching prove to be superior estimators even when given a full budget of steps.

If you wanted the best sample quality regardless of speed, which method would you choose? If you wanted fast generation with acceptable quality, which would you choose?

- **Best Quality:** I would choose **DDIM at 1000 steps**. In our experiments, it achieved the absolute lowest KID score (0.0020), producing the most faithful samples.
- **Fast Generation:** I would choose **Flow Matching at 50 steps**. It achieves a very high-quality KID of 0.0057—which is nearly identical to the "gold standard" DDPM 1000-step baseline—but requires 20x less compute.

Part V: Track Selection (10 points)

For the rest of the course, you'll specialize in one of three tracks. Your choice determines what you'll implement in HW3 and HW4.

The Three Tracks:

- **Fidelity**: Best image quality

  Your goal is to generate the highest quality images possible. This track is all about pushing the limits of sample quality through architecture improvements, better training recipes, advanced samplers, or any other techniques you can find. You'll measure success primarily through KID/-FID scores, but visual quality matters too. Example directions include: trying different model architectures (e.g., DiT, U-ViT), experimenting with training techniques (e.g., noise schedule tuning), or implementing advanced samplers.

  Evaluation Metric: KID/FID ↓

- **Controllability**: User control over generation

  Your goal is to control what the model generates. This is the most open-ended track: you have creative freedom to define what "control" means to you. Some ideas: conditioning on attributes (e.g., generate smiling faces, faces with glasses), text-to-image generation, image editing, interpolation in latent space, style transfer, or anything else you can imagine. Be creative! But remember: with great power comes great responsibility. Since this track is open-ended, you'll also need to define your own evaluation metrics that make sense for your chosen form of control.

  Evaluation Metric: You define it! (e.g., CLIP score, attribute classifier accuracy, user study, or custom metrics that fit your approach)

- **Speed**: Fast inference

  Your goal is to generate good images in as few sampling steps as possible. This track explores the tradeoff between speed and quality: can you match your baseline quality with 10x fewer steps? 100x fewer? Or even with single step sampling? You'll explore techniques like distillation,

consistency models, better ODE solvers, or learned step-size schedules. Success is measured by how few steps you need to reach a target KID threshold.

Evaluation Metric: Number of steps to reach KID $<$ threshold (e.g., KID < 0.005)

Q6. Track Selection [10 pts]

(a) [2 pts] Which track do you choose?

Controllability! No uses generating stuff if you cannot control the output.

(b) [3 pts] Why does this track interest you? What draws you to this particular challenge?

As I stated, sampling from distributions is fine, but diffusion models find applications in a broad scope beyond image generation. For things like robot trajectory generation with flow matching (a very active research field), having some semblance of control over the output is very important. I want to investigate these applications in my own research as well, hence controllability is the way to go.

(c) [3 pts] What's one thing you're uncertain about or curious to explore in your chosen track?

Text to image generation. I am curious to see how accurately I can control the generation of images with textual prompting. Modern methods like Nano-Banana are quite good at this, but still occasionally goof-up. Understanding them better would be useful.

(d) [2 pts] What resources (papers, repos, tutorials) have you found that might help with your chosen track? List at least 2.

1. StableDiffusion: High-Resolution Image Synthesis with Latent Diffusion Models (Rombach et al.)
2. T2I-Adapter: Learning Adapters to Dig out More Controllable Ability for Text-to-Image Diffusion Models (Mou et al.)
3. GLIGEN: Open-Set Grounded Text-to-Image Generation (Li et al.)

Part VI: Reflection (5 points)

Q7. Reflection [5 pts]

(a) [2 pts] Looking back at HW1 and HW2, what's the most surprising or interesting thing you learned about diffusion models and flow matching? What intuition did you build that you didn't have before?

For someone like me with a Physics background, flow matching is more intuitive. I was surprised at the simplicity of the idea and how easy it is to implement. DDPM seemed like a hit-or-miss strategy, while Flow matching seems more natural.

(b) [2 pts] Based on your experience with DDPM, DDIM, and Flow Matching: if you were starting a new generative modeling project today, which method would you choose as your starting point? Why?

I think I answered this pretty much in the previous question, but definitely Flow Matching. It is fast, easy to use, and intuitive. It seems like a good starting point and something that can be easily extended.

(c) [1 pts] List all the resources you used for this homework: AI tools, open source code, tutorials, papers, classmates, etc.

Academic Papers

1. Lipman et al. (2023): Flow Matching for Generative Modeling.
2. Song et al.(2020): Denoising Diffusion Implicit Models.
3. Ho et al. (2020). Denoising Diffusion Probabilistic Models.

Code bases and Blogs referenced

1. TorchCFM (Conditional Flow Matching) Repository on GitHub.
2. Official PyTorch Implementation of Flow Matching (Facebook Research).

AI Helpers

1. **Google Gemini** - Primary coding assistant. Used for implementing the Flow Matching `compute_loss` and Euler ODE solver, debugging Modal cloud deployment, generating the automated ablation script, and formatting LaTeX tables.
2. **VS Code Copilot** - Auto-completion and syntax corrections.

That's it! You have completed HW 2! Congrats on your freshly baked diffusion models!

References

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851, 2020.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow

matching for generative modeling. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=PqvMRDCJT9t.

Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.